

BE 530 Final Project
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Part I — Classical Machine Learning

1. Problem framing and dataset suitability

I selected the medical “Retinal OCT images dataset” from online source:

<https://www.kaggle.com/datasets/paultimothymooney/kermany2018>

I created smaller dataset using 1000 OCT images for each 4 disease folders inside of “train” folder while “test” and “val” folders I kept untouched

- **The task** is a multi-class classification problem where retinal OCT images must be classified into one of four disease categories: CNV, DME, DRUSEN, NORMAL

The goal is to develop model capable of automatically identifying retinal diseases from medical images

- **The unit of analysis** is one OCT image representing a retinal scan from a patient. Each image is treated as an independent data sample

- The **input** After preprocessing the model uses its pixel intensity values of resized image as numerical features. The raw input consists of grayscale OCT images

Images were transformed into feature vectors using the following steps:

- resize images to 128×128 pixels
- convert to grayscale
- normalize pixel values to $[0, 1]$
- flatten images into 1-dimensional feature vectors

- the **target/label**

The dataset contains four classes:

- CNV – Choroidal Neovascularization
- DME – Diabetic Macular Edema
- DRUSEN – associated with age-related macular degeneration
- NORMAL – Healthy retina

- The **baseline comparator** helps evaluate whether more advanced models improve performance. A most frequent class classifier was used as a baseline

Baseline results: Validation accuracy = 0.25 Test accuracy = 0.25

This represents random performance for a balanced 4-class classification problem

- **The Retinal OCT dataset is suitable** for this part because it is:

- public biomedical dataset
- it contains labeled medical images
- it has a large number of samples, that supports machine learning training
- it represents a real clinical task (retinal disease detection)
- it allows both classical machine learning and deep learning approaches

Risks in the dataset:

1 Leakage may occur if images from the same patient appear in training and in test folders.

The train/validation/test split should be unmixed to prevent leakage. Also preprocessing steps such as scaling and PCA were fit only on the training data

2 Confounding

Image acquisition conditions (different devices or hospitals) can influence image appearance and potentially bias the model. OCT images may contain patterns unrelated to disease but related to imaging conditions

3 Class imbalance

Some disease classes contain more images than others, which can bias the model and defend majority classes. Class imbalance is not present in our dataset because each class contains the same number of samples

4 Label ambiguity

Some retinal diseases exhibit similar visual characteristics, which may cause ambiguity in classification.

5 Unrealistic framing. Real clinical deployment would require larger datasets and external validation

2. Preprocessing and Evaluation Design

Train / validation / test strategy

The training set contained 4000 selected images (1000 per class), I limited the set to max 300 images. So the training set contained 1200 selected images (300 per class), the validation set contained 32 images (8 per class), and the test set contained 968 images (242 per class).

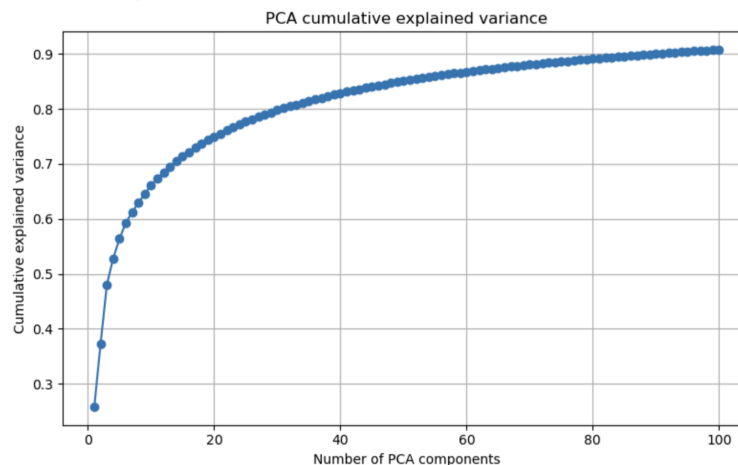
Training data was used to train models, validation data was used for model tuning, and test data was used for final evaluation

```
Train: 1200
Val: 32
Test: 968
Train: Counter({'CNV': 300, 'DME': 300, 'DRUSEN': 300, 'NORMAL': 300})
Val: Counter({'CNV': 8, 'DME': 8, 'DRUSEN': 8, 'NORMAL': 8})
Test: Counter({'CNV': 242, 'DME': 242, 'DRUSEN': 242, 'NORMAL': 242})
```

Preprocessing steps

Each OCT image was converted to grayscale, resized to 128×128 pixels, converted to grayscale, pixel normalized to the range $[0,1]$ and This produced 16384 features per image. It is extremely high so Principal Component Analysis (PCA) was applied to reduce dimensionality from 16384 features to 100 principal components. The first 100 components explain approximately 90.7% of the total variance.

```
Classes: [np.str_('CNV'), np.str_('DME'), np.str_('DRUSEN'), np.str_('NORMAL')]
Scaled train shape: (1200, 16384)
Scaled val shape: (32, 16384)
Scaled test shape: (968, 16384)
PCA train shape: (1200, 100)
PCA val shape: (32, 100)
PCA test shape: (968, 100)
```



Explained variance by 100 components: 0.90768874

Leakage prevention

To prevent information leakage, StandardScaler and PCA were fit on the training set only. The same fitted transformations were then applied to validation and test sets.

Accuracy was used as a simple overall performance metric because the final selected dataset was class-balanced across classes in the reduced subset. In addition, a confusion matrix was used to analyze class-specific errors and identify which disease categories were confused most often.

Evaluation metrics

The following metrics were used:

- accuracy
- precision
- recall
- F1-score
- confusion matrix

3. Baseline and model comparison

A majority-class dummy classifier was used as the baseline. Since the classification task has four classes, baseline accuracy is approximately 25%, corresponding to random or most-frequent prediction under balanced class sampling.

Three classical machine learning models were implemented:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest

Among the evaluated classical machine learning models, SVM achieved the highest test accuracy (about 41.3%), while Random Forest achieved the highest validation accuracy (about 36.6%). Logistic Regression performed close to baseline (35.6%), suggesting that the classification problem is not easily separated by a simple linear decision boundary after PCA compression

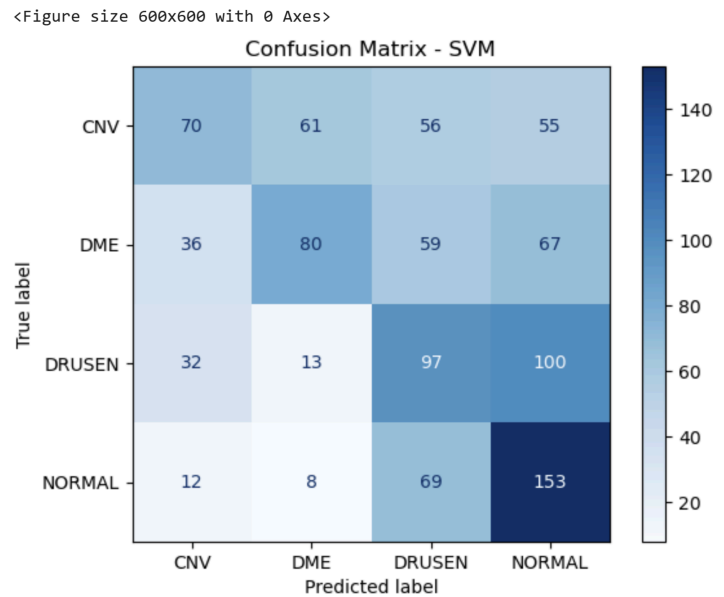
```
Logistic Regression
Validation accuracy: 0.25
Test accuracy: 0.35640495867768596
```

```
SVM
Validation accuracy: 0.3125
Test accuracy: 0.4132231404958678
```

```
Random Forest
Validation accuracy: 0.40625
Test accuracy: 0.36673553719008267
```

Confusion matrix:

The confusion matrix shows that the SVM model performed best on the NORMAL class, while CNV, DME, and DRUSEN were more frequently confused with one another. This suggests that disease-related OCT patterns overlap and are difficult to separate using classical machine learning on flattened image features.



4. Results and interpretation

The results indicate that classical machine learning methods achieve moderate performance with a maximum accuracy of approximately 41%. Flattening images into feature vectors removes spatial information, which likely reduces classification performance.

	precision	recall	f1-score	support
CNV	0.47	0.29	0.36	242
DME	0.49	0.33	0.40	242
DRUSEN	0.35	0.40	0.37	242
NORMAL	0.41	0.63	0.50	242
accuracy			0.41	968
macro avg	0.43	0.41	0.41	968
weighted avg	0.43	0.41	0.41	968

5. Error Analysis and Limitations

The model failed mainly by confusing disease categories with one another, especially CNV, DME, and DRUSEN. This is visible in the confusion matrix, where off-diagonal entries remain high. One likely reason is that classical machine learning models operate on flattened pixel features and therefore do not explicitly capture the spatial structure of retinal lesions. Another limitation is the very small validation set (8 images per class), which may make validation estimates unstable. In addition, resizing images to 128×128 simplifies computation but may remove fine retinal details.

The best-performing classical model was SVM, but overall test accuracy remained moderate. Therefore, the results justify the claim that classical machine learning can extract some diagnostic signal from OCT images, but they do not justify strong claims of clinical usefulness. A deep learning CNN is likely to perform better because it can learn spatial image patterns directly.

In addition to overall accuracy, class-wise precision, recall, and F1-score were computed to better understand model performance for each retinal disease category.

Model performance

The baseline majority-class classifier achieved an accuracy of 25%, which corresponds to random guessing in a four-class classification problem.

Among the evaluated classical machine learning models, the Support Vector Machine (SVM) achieved the highest test accuracy of approximately 41%. Random Forest performed similarly but slightly worse on the test set, while Logistic Regression performed close to baseline.

This suggests that the relationship between features and labels is not well captured by a simple linear decision boundary.

The classification report

The classification report shows that the NORMAL class achieved the highest recall (0.63), indicating that healthy retinal images were easier for the model to identify. In contrast, CNV and DME had lower recall values, suggesting that these disease classes were often confused with other categories. This pattern is consistent with the confusion matrix, where disease categories show higher misclassification rates.

Model	Validation Accuracy	Test Accuracy
Baseline	0.25	0.25
Logistic Regression	0.25	0.36
SVM	0.31	0.41
Random Forest	0.41	0.37

SVM achieved the best test performance and was therefore selected for detailed error analysis.

Error Analysis:

The moderate accuracy suggests that classical machine learning models using flattened pixel features cannot fully capture the spatial structure of OCT images

The pipeline is

OCT image → Grayscale conversion → Resize 128×128 → Flatten (16384 features) → Standardization → PCA (100 components, 90.7% variance) → Baseline → Logistic Regression → SVM → Random Forest → Evaluation → Confusion matrix → Classification report

The moderate accuracy likely results from the limitations of the feature representation. Flattening images into pixel vectors removes spatial relationships between pixels. As a result, classical machine learning models may struggle to distinguish subtle retinal patterns.

Classical machine learning approaches have limited ability to capture complex spatial structures in OCT images

Where the model failed

Errors occurred mainly between disease classes:

-CNV vs DME

-DRUSEN vs NORMAL

Reasons for failure

Possible causes include:

- loss of spatial structure after flattening images
- similarity between retinal disease patterns
- limited dataset size

Potential improvements

Future improvements could include:

- using convolutional neural networks
- applying data augmentation
- using transfer learning

Part II — Deep Learning

I selected the same medical “Retinal OCT images dataset” from online source:

<https://www.kaggle.com/datasets/paultimothymooney/kermany2018>

1. Problem framing and dataset suitability

The same OCT dataset was used for deep learning classification.

Deep learning is appropriate because CNNs can automatically learn spatial features from image data

2. Model rationale and training design

Model architecture

A Convolutional Neural Network (CNN) was implemented using TensorFlow.

Architecture:

Input: 128 × 128 grayscale image

Layers:

Rescaling

Conv2D (32 filters)

MaxPooling

Conv2D (64 filters)

MaxPooling

Conv2D (128 filters)

MaxPooling

Flatten

Dense (128)

Dropout

Dense (4 output classes)

Total parameters: 3,304,580 trainable parameters

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 128, 128, 1)	0
conv2d (Conv2D)	(None, 126, 126, 32)	320
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516

Total params: 3,304,580 (12.61 MB)

Trainable params: 3,304,580 (12.61 MB)

Non-trainable params: 0 (0.00 B)

Preprocessing Images were:

- resized to 128×128
- normalized
- loaded using TensorFlow datasets
- batched for training

Loss function

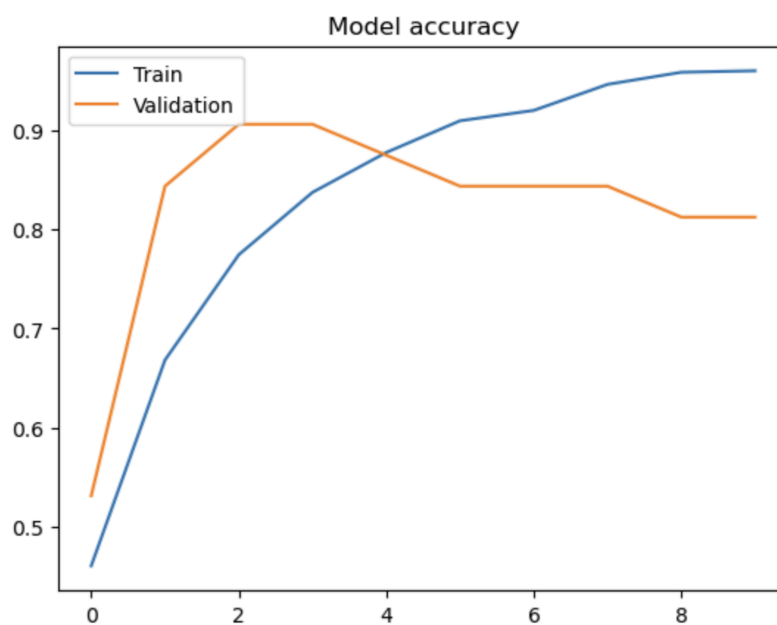
The model used “Sparse categorical cross-entropy” It is appropriate for multi-class classification

Optimizer: The Adam optimizer was used for training.

Regularization Dropout was included in the dense layer to reduce overfitting

3. Results and interpretation

Learning curves



Interpretation:

Training accuracy increases steadily while validation accuracy stabilizes around 0.81–0.84, indicating good learning performance with mild overfitting

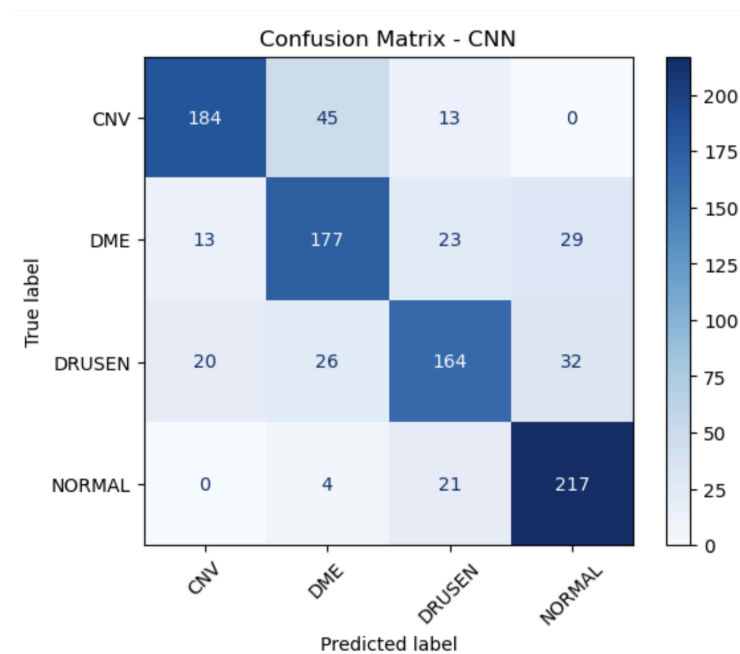
Final test performance

Test accuracy: 76.6%

Classification report: the table showing precision, recall, and F1-score

	precision	recall	f1-score	support
CNV	0.85	0.76	0.80	242
DME	0.70	0.73	0.72	242
DRUSEN	0.74	0.68	0.71	242
NORMAL	0.78	0.90	0.83	242
accuracy			0.77	968
macro avg	0.77	0.77	0.77	968
weighted avg	0.77	0.77	0.77	968

Confusion matrix



Interpretation:

The confusion matrix shows that the CNN model correctly classifies most images across all classes, with the highest performance for the NORMAL class. Some confusion occurs between CNV, DME, and DRUSEN due to similarities in retinal structure patterns

Overall results:

- * Test accuracy: ~0.77
- * Macro F1-score: ~0.77
- * Best class: NORMAL
- * Most confusion: between disease classes

4. Error analysis and limitations

Failure cases: Some DRUSEN images are misclassified as NORMAL or DME due to similar visual structures.

Signs of overfitting

Training accuracy approaches 96% while validation accuracy stabilizes around 81–84%, indicating mild overfitting

Justification of architecture

The CNN architecture effectively captures spatial patterns in retinal images

Limitations

Limitations include:

- relatively small training dataset
- limited validation set size
- absence of data augmentation

Part III — Comparative Synthesis

1. In this project I used one dataset of retinal OCT images for both the classical machine learning and deep learning parts because using the same dataset allowed me to see a fair comparison between the different models. It was a very interesting experience for me. This makes easier to see how much improvement deep learning provides compared to classical machine learning

2. One dataset was suitable for both parts because the OCT retinal dataset contains labeled medical images of four retinal conditions: CNV, DME, DRUSEN, and NORMAL.

For the classical machine learning model the images were converted into numerical feature vectors by resizing them, converting them to grayscale, normalizing the pixel values, and flattening the images into vectors with 16,384 features. Then the PCA was applied to reduce the dimensionality to 100 components with keeping about 90.7% of the variance

For the deep learning part the original image structure was preserved and used directly by CNN. It automatically learns patterns from images.

It works well for both because the dataset contains image data with clear labels.

3. I used one

4. Deep learning provided significantly better performance compared to classical machine learning models

The best classical machine learning model was the Support Vector Machine which achieved test accuracy about 41%. Logistic regression achieved about 36% and the random forest achieved about 37%

In comparison the convolutional neural network achieved a test accuracy about 77% which is almost double of the classical models performance

This improvement happens because CNN models can learn spatial features from images but classical machine learning models rely on flattened image vectors that lose important spatial features

5. The additional complexity of the CNN model was justified because it significantly improved the performance.

The CNN model contains around 3.3 million trainable parameters, which makes it more complex than classical machine learning models. This complexity allows the model to learn detailed visual patterns in the retinal images

The large improvement in accuracy from about 41% to 77% shows that the added complexity is worthwhile for this type of image classification problem. But the CNN requires more computational resources and longer training time,

6. Claims across the full project

Based on the results it concluded: deep learning models perform better than classical machine learning models for image classification tasks, especially if data contains complex visual patterns

The CNN model achieved higher accuracy and better precision and recall scores across all “disease classes” compared to the classical ML models.

But my claims cannot be fully supported because the dataset I used in this project is relatively small and the models were tested only on this dataset. The results may not generalize to other retinal datasets or real clinical environments without further validation

My future works could include training on larger datasets, using augmentation and testing the model on external datasets to confirm its reliability.